

SUMMARY

AI systems engineer with 6+ years of experience building multimodal vision models, production ML training/serving infrastructure, and open-source model integrations in Hugging Face Transformers. Experienced in TensorRT/Triton inference optimization, model conversion, efficient training, and performance-oriented ML systems.

EXPERIENCE

- Toss Bank** Seoul, South Korea
Data Scientist Jan 2026 - Present
 - **On-premise AI Agent System:** Building an on-premise opencode agent system for secure internal AI-assisted development workflows.
 - **AI Authentication:** Developing AI-based authentication systems for face and ID card verification.
 - **Document AI:** Developing end-to-end document information extraction systems using vision-language models for banking workflows and identity AI pipelines.
- SuperbAI** Seoul, South Korea
Machine Learning Engineer Sep 2021 - Jan 2026
 - **[Lead] Vision Foundation Model:** Led development of a multi-modal vision foundation model for text/semantic-prompted object detection and segmentation; prepared a 1.1M-image in-house dataset and drove planning through API deployment and AWS Marketplace release.
 - **CVPR 2025 Competitions:** Won 2nd place in IOD and 4th place in FSOD challenges, validating the model on open evaluation benchmarks.
 - **[Lead] ML Training & Serving Platform:** Built AWS Batch based training infrastructure and optimized production inference with TensorRT and Triton, improving throughput by 5x versus pure PyTorch serving; generated 1,100+ models and 60+ endpoints in one year.
 - **Efficient Training:** Reduced GPU memory usage by 65.6% and training time by 44.3% using parameter- and memory-efficient techniques while preserving model performance.
 - **Interactive Segmentation:** Built browser, hybrid, and cloud-based segmentation tools using RepViT-SAM, FocalClick, SAM, and SAM2, increasing segmentation labeling speed by 25x.
- Kakao Enterprise** Pangyo, South Korea
Machine Learning Engineer Intern Feb 2021 - Aug 2021
 - **AutoML:** Fixed AutoGluon NeuralNetFastAI scaling issue and developed a Flask-based training/inference AutoML framework with a simple front-end.
- Team ISLAND** Seoul, South Korea
Chief Technology Officer Mar 2019 - Oct 2020
 - **Startup:** Raised approximately \$290K through government subsidies and seed funding; managed five developers across Android, Unity, and deep learning workstreams.
 - **Mobile ML:** Developed a lightweight 3D human pose estimation model for mobile devices and converted PyTorch models to TensorFlow Lite.

OPEN SOURCE

- **Hugging Face Transformers:** Contributed 40+ pull requests across multimodal, vision, and object-detection models, including SAM2, Molmo2, RT-DETR, ViTPose, DINOv3 utilities, and training/inference fixes.
- **Hugging Face Transformers - SAM2:** Led the addition of Segment Anything 2 support to Transformers, including model integration, image/video processors, conversion flow, documentation, and tests; credited in the official [SAM2 docs](#).
- **Hugging Face Transformers - Molmo2:** Opened Molmo2 support for Transformers and published [danelcsb/Molmo2-4B](#) on the Hugging Face Hub.

PAPERS

- **ZERO: Multi-modal Prompt-based Visual Grounding:** arXiv - Jul 6, 2025
- **MobileHumanPose: Toward real-time 3D human pose estimation in mobile devices:** IEEE/CVF CVPR Workshop - Jun 19, 2021
- **Improved DOA estimation of MUSIC algorithm based on irregular microphone array:** UIUC Research - Dec 20, 2018
- **On Joint Transfer of Energy and Information: A Markov Decision Problem Formulation:** Research paper - Dec 13, 2018
- **Terahertz Signal Analysis in Tissue of Gastroesophageal Reflux Disease:** Sejong Science High School - Dec 31, 2013

EDUCATION

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| • KAIST (Korea Advanced Institute of Science and Technology)
<i>Professional Master in Electrical Engineering; GPA: 3.55/4.3</i> | Daejeon, South Korea
<i>Feb. 2020 – Aug. 2021</i> |
| • POSTECH (Pohang University of Science and Technology)
<i>Bachelor in Electrical Engineering; GPA: 3.4/4.3</i> | Pohang, South Korea
<i>Mar. 2015 – Feb. 2020</i> |
| • UIUC (University of Illinois Urbana-Champaign)
<i>Exchange Student; GPA: 3.6/4.0</i> | Illinois, United States
<i>Jan. 2018 – Dec. 2018</i> |

PROGRAMMING SKILLS

- **AI Systems:** Python, PyTorch, Transformers, TensorRT, Triton, AWS, MLflow **Experienced:** TensorFlow, model conversion, Swift, LangChain, Langfuse, Claude Code, Codex, opencode